

HAREN GANNARAPU

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EDUCATION

University of Central Florida

Bachelor of Science in Computer Science

- **GPA: 3.8** | Dean's List (All Semesters) | Bright Futures Scholar
- Coursework: Data Structures & Algorithms, Operating Systems, Systems Software, Object-Oriented Programming, NLP
- Activities: **AI@UCF** — Vice President (Fall 2026), **KnightHacks** — Operations Team

Orlando, FL

Expected Dec 2027

EXPERIENCE

Software Engineering Intern

Nebo Agency

- Build mobile-first features across **100 components** on **2 retainer clients'** live WordPress sites (PHP, SCSS, jQuery).
- Run both sites locally in Docker (PHP 8.2, MariaDB); traced a new-developer setup failure to an unset Docker Compose variable.
- Scope changes in recurring client calls and ship them through GitKraken and GitHub with Gulp builds and staging review.

Aug 2026 – Present

Remote

Software Engineering Intern

Carnival Corporation (Princess Cruises)

- Built a Retrieval-Augmented Generation (RAG) assistant for **100+ engineers**: wrote the Python retrieval service grounding **Azure OpenAI** answers in Jira and Confluence, the **Spring Boot API** on Azure, secret redaction, and hallucination tests.
- Architected an event-driven agentic pipeline on **Atlassian Rovo** that matches new Jira bugs to past fixes by root cause and auto-drafts Confluence runbooks; **14 adversarial test tickets** caught 4 bugs, incl. 2 redaction failures.
- Built a Microsoft Copilot Studio agent that queries live Jira and Confluence from Teams over Atlassian's Model Context Protocol (MCP) server with secret redaction, after a 68-bug audit found only **1 in 8** past fixes reusable.

Jun 2026 – Aug 2026

Fort Lauderdale, FL

Machine Learning Engineering Intern

Oasis (AI Travel Startup)

- Cut the nightly ML pipeline from **45 to 8 minutes** by swapping full PostgreSQL dumps for incremental loads and S3 caching.
- Built a heterogeneous GraphSAGE graph neural network (GNN) in PyTorch Geometric for link-prediction recommendations on a **2M-edge** user-post-tag graph, served in real time via NVIDIA Triton.
- Configured KEDA autoscaling on SQS queue depth to scale Triton inference pods on Amazon EKS from **2 to 20 replicas**, matching GPU capacity to traffic instead of provisioning for peak.
- Caught a tokenizer regression adding **120ms to p99 latency** within hours of deploy via Prometheus and Grafana monitoring.

May 2025 – Aug 2025

PROJECTS

Paradise — *ShellHacks 2026* | *Node.js, ONNX Runtime, YOLOv10, Grounding DINO, CLIP, Depth Anything, WebHID* Sep 2026

- Won **3rd Place Overall** at ShellHacks 2026, Florida's largest hackathon, among **300+ projects** and **1,400+ hackers**.
- Built touch-only navigation for DeafBlind users that walks them to a Waymo and steers their hand onto its door handle through Joy-Con haptics, fusing **5 computer vision models** across a chest phone and laptop in a **50ms** guidance loop.
- Trained a Waymo classifier (logistic regression on frozen CLIP ViT-B/32 embeddings) on **779 vehicle crops** incl. look-alike robotaxis, reaching **~90% recall** with **4 false positives in 654** negatives under 5-fold grouped cross-validation.
- Converted relative Depth Anything V2 output into metric obstacle stops at **80ms/frame** with a RANSAC floor-plane fit, and cut open-vocabulary detection from **14s to 1.2s/frame** by moving Grounding DINO to ONNX Runtime workers.

ReplAI — *Hack The Bay 2026* | *Next.js, TypeScript, Gemini 2.5 Flash, LiveKit, MediaPipe, Three.js*

Apr 2026

- Won **1st Place** in the **Zenardy Challenge** and **Best Use of Gemini**, and placed **3rd Overall of 34 projects**.
- Held speech-to-feedback latency under **300ms** in a browser conversation coach fusing Gemini 2.5 Flash dialogue, MediaPipe Face Mesh eye-contact tracking, and LiveKit prosody analysis over a shared event emitter with LLM backpressure.
- Flagged adverse parties with **sub-second** detection via a real-time conflict-of-interest pipeline that used Named Entity Recognition (NER) on live transcripts and matched parties to a firm database with a hybrid Levenshtein + Jaccard scorer.

Revive — *HackaBull VII 2026* | *Next.js, TypeScript, Arduino, FSR 406, Gemini, ElevenLabs, Web Serial*

Apr 2026

- Won **Best Use of ElevenLabs** at HackaBull VII among **58 competing projects**.
- Suppressed **30%+** of false compression counts in a CPR coaching device by streaming **200Hz** force-sensor data over Web Serial into a ring buffer with dual-threshold hysteresis in Arduino firmware.
- Held compression-to-coaching latency under **2 seconds** by orchestrating **3 specialized Gemini agents** (patient physiology, AHA-target evaluation, scenario progression) behind an event bus batching every **500ms**.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, PHP, C, C++, SQL, HTML/CSS (SCSS)

AI/ML: PyTorch, PyTorch Geometric, Large Language Models (LLMs), RAG, Computer Vision, ONNX Runtime, Hugging Face

Frameworks: Spring Boot, React, Next.js, Node.js, Express, FastAPI, WordPress, jQuery, Three.js

Cloud & DevOps: AWS (EKS, S3, EC2, RDS, SQS, VPC, IAM), Azure, Kubernetes, Docker, Terraform, CI/CD (GitHub Actions)

Tools: Git, GitKraken, Linux, PostgreSQL, REST APIs, Prometheus, Grafana, Jira, Confluence, Model Context Protocol (MCP)